

JGR Atmospheres

RESEARCH ARTICLE

10.1029/2024JD042442

Key Points:

- Satellite reentry material accumulates at high latitudes between 10 and 30 km
- Small perturbations to heating rates results in as much as 1.5 K temperature anomalies at high latitudes in the middle atmosphere
- Temperature anomalies caused by satellite reentry may impact the stratospheric polar vortex and thus stratospheric ozone

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

C. M. Maloney,
christopher.maloney@noaa.gov

Citation:

Maloney, C. M., Portmann, R. W., Ross, M. N., & Rosenlof, K. H. (2025). Investigating the potential atmospheric accumulation and radiative impact of the coming increase in satellite reentry frequency. *Journal of Geophysical Research: Atmospheres*, 130, e2024JD042442. <https://doi.org/10.1029/2024JD042442>

Received 10 SEP 2024

Accepted 6 FEB 2025

Author Contributions:

Conceptualization: Christopher M. Maloney, Robert W. Portmann, Martin N. Ross, Karen H. Rosenlof
Data curation: Christopher M. Maloney
Formal analysis: Christopher M. Maloney
Funding acquisition: Christopher M. Maloney, Karen H. Rosenlof
Investigation: Christopher M. Maloney, Robert W. Portmann, Martin N. Ross
Methodology: Christopher M. Maloney
Project administration: Karen H. Rosenlof
Resources: Martin N. Ross
Software: Christopher M. Maloney

© 2025. The Author(s). This article has been contributed to by U.S. Government employees and their work is in the public domain in the USA.

This is an open access article under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Investigating the Potential Atmospheric Accumulation and Radiative Impact of the Coming Increase in Satellite Reentry Frequency

Christopher M. Maloney^{1,2} , Robert W. Portmann² , Martin N. Ross³ , and Karen H. Rosenlof² 

¹Cooperative Institute for Research in Environmental Sciences, Boulder, CO, USA, ²National Oceanic and Atmospheric Administration, Boulder, CO, USA, ³The Aerospace Corporation, El Segundo, CA, USA

Abstract Construction of numerous satellite megaconstellations in the low Earth orbit (LEO) (300–2,000 km) is projected over the coming decades. Estimates suggest that the number of satellites in an LEO could exceed 60,000 by 2040. The increase in the annual mass flux of anthropogenic material into the upper atmosphere as a result of maintaining these megaconstellations could rival the natural occurring meteoric mass flux. Little is known about the aerosols that will be produced by reentry vaporization, which makes estimating the associated impacts on climate and ozone difficult. Aluminum is a primary satellite component that will likely be emitted during reentry vaporization. In this study we simulate a reentry emission of 10 Gg/yr, assuming that all aerosols released is aluminum oxide (Al₂O₃). This level of Al₂O₃ emission is consistent with expected megaconstellation growth by 2040. We investigate how the location of atmospheric accumulation, aerosol size distribution, and radiative properties of reentry Al₂O₃ impacts the middle-to-upper atmosphere. We find that depending on reentry latitude and aerosol size distribution, a 20–40-Gg stratospheric burden of Al₂O₃ aerosols accumulates poleward of 30 N/S between 10 and 30 km. Small but statistically significant changes in mesospheric heating rates lead to 1.5 K-temperature anomalies in the mesosphere and the stratosphere at Southern Hemisphere high latitudes. These temperature anomalies are accompanied by a 10% reduction in wind speed in the Southern Hemisphere polar vortex, leading to a weaker springtime ozone hole. Some reentry scenarios also experience a strengthening of the Northern Hemisphere polar vortex.

Plain Language Summary Large increases in the number of satellite megaconstellations in the low Earth orbit (300–2,000 km) are projected, with the number of satellites in an orbit exceeding 60,000 by 2040. This will lead to annual emissions of man-made material into the upper atmosphere on the same order of naturally occurring emissions from meteorites. A lack of observations and validated models of reentry demise limits our ability to simulate the complex aerosols associated with reentry, which makes estimating the climate impacts difficult. Aluminum is a primary satellite component and will likely be emitted during reentry vaporization in the form of alumina. Unmodified alumina is a useful approximation for metallic reentry aerosol. In this study, we simulate a potential yearly emission of 10,000 metric tons of alumina from reentering space debris. We investigate how the location of atmospheric accumulation, aerosol size distribution, and radiative properties of reentry alumina impacts the middle atmosphere. We find that 20,000–40,000 metric tons of alumina accumulates at high latitudes between 10 and 30 km in both hemispheres. Small changes in mesospheric heating rates lead to 1.5-K temperature anomalies in the middle atmosphere at high latitudes. These temperature anomalies are accompanied by changes in wind speed in the polar vortex.

1. Introduction

The space industry has rapidly grown over the course of the past decade, leading to a more than tripling of annual launch rates (Ailor, 2021; Jones, 2018) with expectation for additional growth in the coming decades (Jones, 2018). Much of the recent scientific investigation in regards to space travel has been focused upon emissions from increased rocket launch rates (i.e., Larson et al., 2017; Maloney et al., 2022; Ryan et al., 2022). However, with a decrease in launch costs, as well as an increase in the capabilities of low Earth orbit (LEO) satellites, large LEO constellations are poised to eclipse rocket launches as the primary middle-to-upper atmosphere emission source in the coming decade. According to recent projections from the U.S. Government Accountability Office, the LEO satellite population from domestic constellations alone are expected to increase to over 60,000 units between 2030 and 2040, a roughly 10 times increase in the current LEO population

Supervision: Robert W. Portmann, Karen H. Rosenlof

Visualization: Christopher M. Maloney, Robert W. Portmann, Martin N. Ross, Karen H. Rosenlof

Writing – original draft: Christopher M. Maloney

Writing – review & editing: Robert W. Portmann, Martin N. Ross, Karen H. Rosenlof

(GAO, 2022). Other estimates suggest that the future population of satellites in LEO could be far larger and approach 1 million in number (Falle et al., 2023).

A typical LEO satellite megaconstellation will consist of tens of thousands of satellites with individual satellites possessing a lifetime of roughly 5 years. As a result, annual disposal and replacement of about 20% of the constellation will be required in order to maintain the constellation, which will cause the spacecraft launch rate to potentially increase by a factor of four and reentry disposal rates to increase by a factor of 10. The primary method for disposal of spacecraft (satellites and upper stages) at the end of life is to reduce orbit altitude until atmospheric drag causes an uncontrolled reentry. Aerodynamic heating during reentry causes surface temperatures of spacecraft structures to exceed their melting point (1,200–3,000 K), resulting in the vaporization of most of the spacecraft (Park et al., 2017). Spacecraft are typically disposed of (a) in an uncontrolled deorbit without regard to geographic location, or (b) in a controlled manner at a specific time and location that minimizes safety hazards. Uncontrolled reentry is not entirely random as the geographic location most likely occurs at latitudes just below the spacecraft orbit inclination (Pardini & Anselmo, 2024). While the latitude dependence of future reentries depends on the unknown orbital parameters of future satellite constellations, latitude bands that are representative of likely future reentry scenarios can be specified. We explore both uncontrolled reentry disposal and targeted disposal into specific latitude bands in this study.

The present-day yearly mass flux of particulate matter released during satellite reentry is approximately 2.0 Gg/yr (Barker et al., 2024), considerably smaller than the estimated annual natural mass flux from meteoric ablation. For context, the annual global ablated meteoric mass flux is estimated to fall between 10 and 20 Gg/yr (i.e., Gardner et al., 2014; Plane, 2012). As the space industry expands, total emissions from the continual end of life disposal of satellites in large LEO constellations is estimated to increase satellite reentry emissions to ~15 Gg/yr by 2040 (i.e., GAO, 2022; Pardini & Anselmo, 2020), which is on the same scale as the annual meteoric mass flux.

The exact chemical and physical make-up of aerosols released during satellite reentry is not well known. Metal and metal oxide mixtures, which may contain iron (Fe), copper (Cu), nickel (Ni), silicates ($[\text{SiO}_{2+n}]^{2n-}$), lithium (Li), and aluminum (Al), among other compounds, are likely produced during reentry (Schulz & Glassmeier, 2021). Recent in situ observations over the northern high latitudes during the Alaska leg of the 2023 Stratospheric Aerosol processes, Budget and Radiative Effects (SABRE) aircraft campaign observed trace amounts of aluminum, copper, lead, and lithium mixed with stratospheric sulfate aerosols. These mixed aerosols appear to be the result of satellite ablation (Murphy et al., 2023).

Aluminum makes up about 35% of an LEO satellite and upper stage rocket's mass (Schulz & Glassmeier, 2021) and has a lower melting point than typical reentry temperatures so that we can assume aluminum compounds represent the greatest single emission from spacecraft reentry. Abundant ambient oxygen (below 80 km) in the presence of high temperatures leads to a rapid growth of oxide layers on metallic surfaces, leading to the production of metal oxide particles. Models of reentry demise generally predict that nearly all spacecraft aluminum is processed into alumina (Al_2O_3) with a net yield approximately equal to 66% of the spacecraft total mass (Park et al., 2021). Other metallic oxides such as MgAl_2O_3 and TiO_2 account for less than 10% of the emitted mass. Therefore, for a plausible future reentry scenario with a 15-Gg/yr satellite and upper stage reentry net mass flux (e.g., GAO, 2022), we assume an Al_2O_3 production rate of 10 Gg/yr.

While the chemical identity of reentry aluminum is well understood, the microphysics of the alumina particles released from surfaces during ablation are not. Various atomic, chemical, and aerodynamic forces act to determine particle size, and a comprehensive model is not available. Recently, Ferreira et al. (2024) modeled aluminum ablation and oxidation reactions at atomic scales for a smaller reentering satellite at a single altitude (80 km). They found that a relatively small yield (20%) of 4 nm AlO particles was produced. Extrapolation of this result to other altitudes and reentry conditions is uncertain. Additional modeling work on aluminum in ablating meteors found that in the low oxygen environment near the mesopause, aluminum oxidizes to AlO. Further oxidation into Al_2O_3 occurs as the AlO sediments into the more oxygen-rich middle atmosphere (Plane et al., 2021). Overall, the process of metallic aerosol production during spacecraft reentry ablation is complex and not well understood. It depends on the details of reentry trajectory, spacecraft structure, and poorly understood altitude dependent processes. Nevertheless, we can reasonably construct reentry emission scenarios with Al_2O_3 aerosols as the primary emission component and altitude profiles adopted from space debris reentry demise models.

The goal of this work is to understand how a hypothetical Al_2O_3 emission of 10 Gg/yr could impact the radiative balance in the Earth's middle-to-upper atmosphere. Al_2O_3 is reflective to shortwave radiation in the visible window, and absorbs infrared radiation in the 10–30- μm wavelength range. Another major goal is to predict the latitude and altitude where reentry Al_2O_3 will accumulate through the action of the atmospheric circulation. Such a finding can be used to inform the deployment of future airborne sampling studies, as well as provide context for both satellite operators and policy makers when considering space debris disposal methods. Five separate simulations were compared to a control run. Investigated in these simulations were four different reentry latitude regions and two separate aerosol size distributions.

Two important factors regarding the composition of reentry aerosols are not considered in this study. First, Al_2O_3 in the middle atmosphere will likely become mixed with stratospheric sulfate (Murphy et al., 2023). As a result, one might expect that the coated Al_2O_3 may have an impact upon heterogeneous ozone chemistry. For example, Al_2O_3 released from solid rocket motors (SRM hereafter) can alter stratospheric ozone chemistry by providing additional surface area for heterogeneous reactions to occur on (Danilin et al., 2003; Hanning-Lee et al., 1996; Ross et al., 1997). Secondly, the particle production of reentry aerosols is not well understood. It is possible that Al_2O_3 vapor from ablation will adhere onto already existing meteoric dust, coalesce with other reentry metals, or follow both pathways. This mixed aerosol will likely further be coated with sulfates in the stratosphere. The sulfate coating and characteristics of mixed aerosols may alter both the radiative and chemical properties of reentry alumina. Unfortunately, due to a lack of both laboratory and in situ observations, we have little knowledge about these processes to inform the modeling effort presented here.

In addition to the uncertainty surrounding reentry aerosols, our version of the Community Aerosol and Radiation Model for Atmospheres (CARMA, Bardeen et al., 2008; Mills et al., 2008; Toon et al., 1988) is unable to directly interact with the heterogeneous aerosol chemistry code in the parent Whole Atmosphere Community Climate Model version 6 (WACCM6, Gettelman et al., 2019; Marsh et al., 2013). Future model development is underway in order to allow us to represent the more advanced chemistry and possible mixed aerosol characteristics of reentry Al_2O_3 . For simplicity, we assume pure α - Al_2O_3 as a first-order approximation for reentry aerosols.

2. Modeling Methods

The simulations performed in this study employed the WACCM6 model coupled with a version of the CARMA model previously used for meteoric dust studies (Bardeen et al., 2008; Mills et al., 2008). CARMA is a sectional microphysical model that represents aerosol size and mass using a variable number of size bins. This flexibility makes CARMA a powerful tool for representing uncertain aerosol size distributions, such as those produced during satellite reentry. Aerosols initially populate the smallest CARMA size bin. From there, aerosols in each bin are allowed to grow in size and move into larger CARMA bins via coagulation with other alumina aerosol. Aerosols in the largest CARMA size bin is assumed to sediment out of the atmosphere at the end of each model time step. Meteoric dust in the Bardeen et al. (2008) CARMA model was altered to represent Al_2O_3 aerosol mass, density, and radiative properties.

Al_2O_3 was assumed to be fractal in shape using the Weisenstein et al. (2015) treatment for alumina emitted for solar radiation management. A Mie scattering approximation for fractal aggregates introduced by Botet et al. (1997) was also employed. The Mie scattering code uses a wavelength-dependent real and imaginary refractive indices data set for the wavelengths between 0.07 and 600 μm . As mentioned at the end of the introduction section, little is known about the optical properties of alumina produced from ablation, so pure α - Al_2O_3 aerosol is assumed. This assumption allows for two observationally based data sets for α - Al_2O_3 to be combined to form an extensive refractive indices data set. The Toon et al. (1976) data set provides the base framework, while the more detailed Tropic and Thomase (1997) data set is used to fill in wavelength gaps in the Toon et al. (1976) data set. Importantly, this data set encompasses the 0.5–1.0 μm scattering and 10–25 μm absorption windows used in Ross et al. (2014) for SRM emitted Al_2O_3 . A visual representation of this data set is shown in Figure S1 in Supporting Information S1.

The size distribution of aerosols produced during satellite reentry is not well constrained, but it is expected to be in submicron in size (Ferreira et al., 2024; Schulz & Glassmeier, 2021). We use two aerosol size distributions that likely bracket the true size for satellite reentry. The first size distribution starts at 0.01 μm and extends to 1.7 μm in the radius. The second size distribution ranges from 0.1 to 17 μm in the radius, which allows for us to look at the impact of potentially very large reentry particles produced by frictional shearing or very rapid coagulation.

Table 1

Naming Convention and Emission Description for Each Reentry Scenario

Reentry scenario	Latitude of emissions	Aerosol emission size (μm)	Annual stratospheric Al_2O_3 burden (Gg)	Stratospheric mmr maximum (ppb)	TOA net flux anomaly (mW/m^2)
RS1: Uncontrolled reentry, small aerosol	40 N/S–80 N/S	0.01	35.0	1.42	-9.62 ± 15.7
RS2: Uncontrolled reentry, large aerosol	40 N/S–80 N/S	0.1	23.3	0.24	1.87 ± 15.7
RS3: Controlled reentry, equator	20 S–20 N	0.01	37.4	1.50	-7.98 ± 15.1
RS4: Controlled reentry, polar	90 S–80 S	0.01	33.5	21.9	-5.50 ± 14.8
RS5: Controlled reentry, South Pacific Ocean Uninhabited Area (SPOUA)	46 S–41 S	0.01	38.6	2.81	-15.0 ± 14.6

Note. The three rightmost columns also list the annual stratospheric alumina burden, maximum mass mixing ratio (mmr), and top of the atmosphere radiative forcing anomaly, respectively. The TOA net flux error bounds represent the model annual variability.

CARMA was configured with 21 size bins for this study. The smallest size bin represents the initial emission radius for each reentry scenario (i.e., 0.01 μm or 0.1 μm). Each subsequent bin increases in size relative to the volume-mass ratio.

Five different reentry scenario perturbation simulations and one control simulation were performed for this study. The scenarios are defined by reentry latitude and aerosol size distribution. Table 1 provides contextual information for each reentry scenario. In each scenario, a constant 10 Gg/yr emission of Al_2O_3 is released between 60 and 70 km. This altitude range is typical for reentry ablation of low melting point metals (Kärräng et al., 2019; Park et al., 2021). Aerosols are emitted at a constant rate throughout the year as the megaconstellation operations model requires continuous disposal of defunct satellites and upper stages used to launch new ones. Four emission latitude bands were defined in order to investigate simulation sensitivity to emission latitude. The emission bands are not necessarily realistic possibilities for all megaconstellation emissions but correspond to different components of different constellations. Scenarios RS1 and RS2 represent uncontrolled reentry disposal in large, midlatitude bands that correspond to disposal from middle inclination orbits that contain the bulk of megaconstellation satellites. RS1 assumes a 0.01- μm size distribution while RS2 assumes a 0.1- μm size distribution. Scenarios RS3 and RS4 represent controlled reentry into the equatorial (20 S–20 N) and polar (80 S–90 S) regions, respectively. RS5 represents controlled reentry into the South Pacific Ocean Uninhabited Area (SPOUA) located between the latitudes of 41 S and 46 S (Ailor, 2022). Scenarios RS3, RS4, and RS5 each use the 0.01- μm size distribution. Due to robust zonal transport in the middle atmosphere, we emit reentry aerosol uniformly in longitude.

Each simulation was run for a total of 35 years using repeating year 2000 conditions for sea surface temperatures, chlorine, and greenhouse gas emissions. The first 5 years of each simulation are excluded in order to allow the middle atmosphere to reach mass equilibrium.

In the results section discussed below, we use the 95% confidence interval to determine statistical significance from the year-to-year model variability. Due to the smaller simulation sample size of 30 years, a bootstrapping method with replacement was used to calculate the statistical significance where the total number of data samples was 1,000.

As previously mentioned, observational uncertainty and modeling limitations make it difficult to simulate the potential stratospheric chemistry involving reentry Al_2O_3 . Thus, reentry aerosol is assumed to be chemically inert and only interacts with the radiation code in these simulations. However, the WACCM middle atmosphere chemistry package is still active in each simulation, which allows for other stratospheric chemistry interactions to occur.

3. Results

3.1. Atmospheric Transport and Accumulation of Reentry Al_2O_3

Most of the simulated reentry scenarios produce an annual Al_2O_3 burden that ranges between 30 and 40 Gg (Table 1). A 30–40 Gg burden from a 10-Gg annual emission is consistent with the approximate 3–5-year lifetime of stratospheric air (i.e., Butchart, 2014; Li et al., 2012; Linz et al., 2017; Waugh & Hall, 2002). RS2 deviates

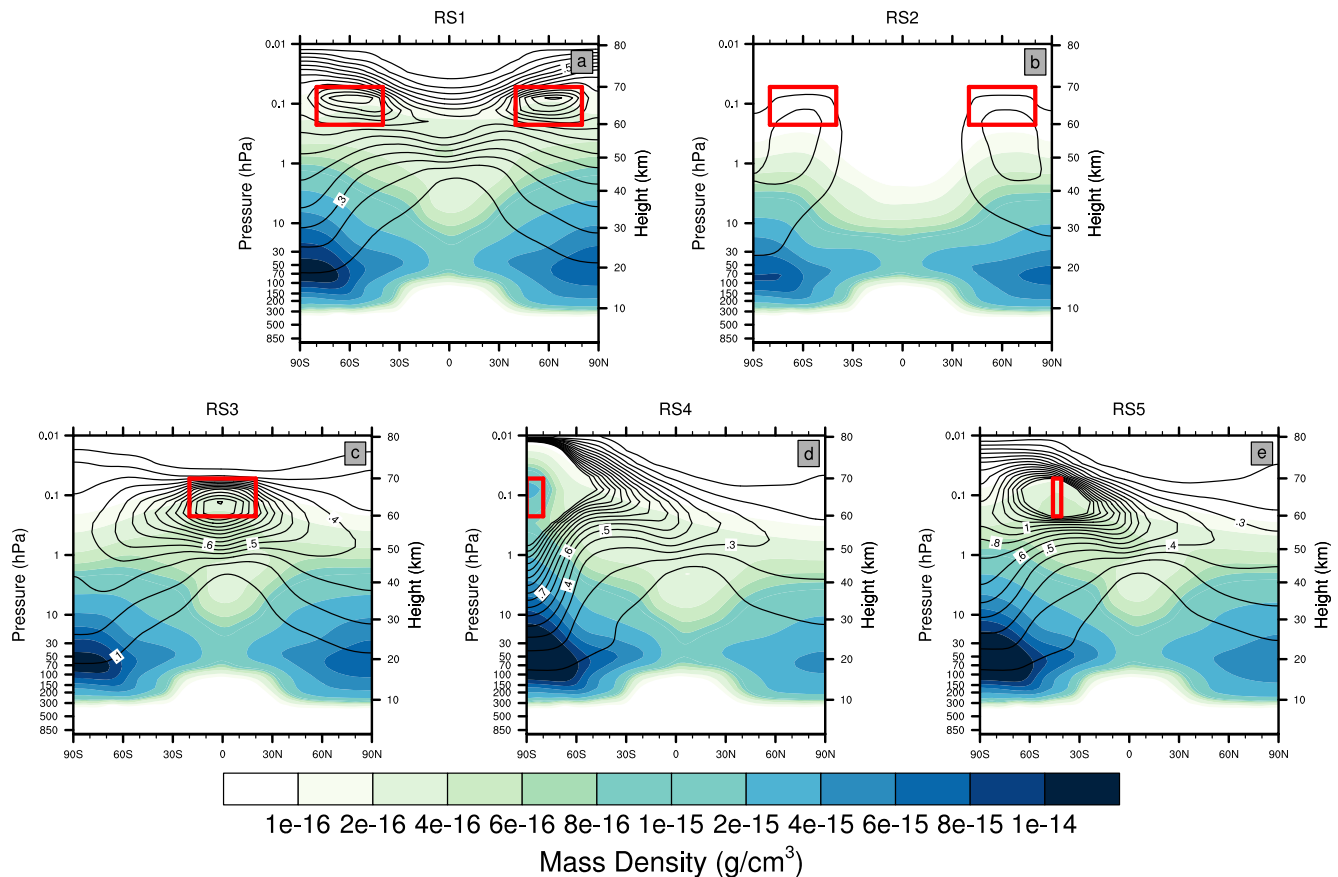


Figure 1. Annual zonal mass density (colored contours) in g/cm^3 and the mass mixing ratio (contour lines) in parts per billion for each reentry scenario. The red box indicates the emission region for each specific scenario.

from the other scenarios as it produces a smaller annual stratospheric burden of 23.3 Gg. The small stratospheric burden in RS2 is a result of the larger aerosol possessing, roughly a 2–3 times more rapid sedimentation speed in the stratosphere than in the other scenarios, shortening its lifetime in the middle atmosphere.

The zonal distribution of Al_2O_3 is further influenced by emission latitude (Figure 1). The mass mixing ratio (mmr) in all scenarios maximizes around each emission region (red boxes, Figure 1), and is listed in Table 1. RS2 produced the smallest maximum mmr (0.24 ppb) while RS4 produced the largest maximum mmr (21.9 ppb). The Al_2O_3 mmr in RS4 and RS5 is mainly confined to the Southern Hemisphere as the seasonality of the middle atmosphere overturning circulation reduces transport from the Southern Hemisphere (SH from hereafter) to the Northern Hemisphere (NH from hereafter) for half of the calendar year (Figures 1d and 1e).

All reentry scenarios that emit the $0.01\text{-}\mu\text{m}$ aerosol show that aerosol mass density accumulates between 10 and 30 km at high latitudes in both hemispheres (Figure 1). The larger-sized aerosol released in RS2 also becomes concentrated at the same region, but the vertical extent is more compact and is roughly 85% and 62% of RS1 in the NH and SH, respectively (Figure 1b). The more symmetrical reentry scenarios have a balanced mass density distribution between hemispheres, while the NH mass density maximum in the asymmetrical reentry cases is roughly 20% of the SH mass density (Figures 1d and 1e).

The annual zonal mass distribution depicted in Figure 1 can in part be explained by variances in the time it takes for aerosols to reach the high-latitude accumulation regions. All reentry emission scenarios begin to amass aerosols in the SH high latitudes within the first 5 months of simulation (not shown); however, the time it takes for the NH aerosol maximum to form varies considerably among each emission scenario (Figure S2 in Supporting Information S1). Using a threshold of one-tenth of the annual mean mass density maximum ($1 \times 10^{15} \text{ g/cm}^3$), we find that the larger aerosol in RS2 takes roughly 8–9 months from the start of the model simulation to exceed the 1/

10th threshold in the NH high-latitude accumulation region (Figure S2 in Supporting Information S1, second row). RS1 and RS3 each reach this 1/10th threshold in late December/early January of the second year (Figure S2 in Supporting Information S1, first and third rows). With its SH polar reentry pattern, RS4 takes the longest to exceed the 1/10th threshold in the NH as the mass density maximum does not form until January of year three (Figure S2 in Supporting Information S1, fourth row). Scenario RS5 is unique as it briefly exceeds the 1/10th threshold for the months of February and March in year two, but then, the change in direction of the overturning circulation blocks additional transport of reentry emissions into the NH during the boreal spring and summer. RS5 once again crosses the 1/10th threshold in November of year two (Figure S2 in Supporting Information S1, fifth row).

It is important to discuss how our findings provide further context to results from the recent Ferreira et al., 2024 satellite demise modeling study. They suggest that it would take a 4-nm-sized AlO particle roughly 30 years to sediment from the ablation region to the ozone layer. However, their study failed to consider the robust transport of air to high latitudes in the mesosphere, and subsequent downwelling into the stratosphere. By including atmospheric circulation in our simulations, we find that the Al_2O_3 will likely reach the altitudes of the ozone layer in a shorter time period of 1–3 years depending on reentry location and aerosol size thus shortening the timeframe for which metallic reentry aerosols may begin influencing stratospheric ozone relative to the Ferreira et al. (2024) estimation.

3.2. Radiative Forcing and Heating of Reentry Al_2O_3

The direct radiative impact of reentry Al_2O_3 is not known. However, earlier work performed by Ross et al. (2014) that estimated the radiative forcing of Al_2O_3 emission from SRM can provide useful context. Ross et al. (2014) assumed that the stratospheric burden of Al_2O_3 was 1.2 Gg prior to the retirement of the solid fuel burning space shuttles. They calculated that the net annual radiative forcing of Al_2O_3 from this 1.2 Gg burden was approximately 4.8 mW/m^2 and was driven by longwave (LW from hereafter) absorption. Additionally, work by Weisenstein et al. (2015) investigated the effectiveness of solid aluminum particles as a hypothetical stratospheric aerosol injection material. They found that aluminum size distribution played an important role in its ability to effectively scatter solar radiation in the atmospheric window ($0.4\text{--}0.7 \mu\text{m}$) with larger aerosols acting as more efficient scatterers.

We are able to estimate the direct radiative forcing of reentry Al_2O_3 by performing a double call to the WACCM6 radiation code. A double call occurs by performing radiation code calculations twice per model time step. The first call executes the radiation calculations normally, while the second call performs the calculations without the influence of aerosols. The difference of these two calculations allows for the estimation of the aerosol radiative forcing. We find that the simulated net TOA radiative forcing (shortwave—longwave) from a 10-Gg/yr reentry alumina flux to be LW driven and on a similar scale as the Ross et al. (2014) findings. RS1, RS3, and RS4 each produce a negative annual net TOA anomaly of 9.62, 7.98, and 5.50 mW/m^2 , respectively (Table 1). RS5 produces a considerably larger negative annual net TOA anomaly of 15.0 mW/m^2 . It is possible that the larger TOA response in RS5 is due to this particular reentry scenario possessing the second largest mesospheric Al_2O_3 mmr (Table 1) and the fact that this mmr maximum persists over a large swath of the low latitude SH, prolonging atmospheric lifetime and ability to absorb LW radiation (Figure 1e). RS5 is the only reentry scenario that possess a net TOA anomaly that exceeds annual model variability. RS2 produces the only positive net TOA radiative forcing of 1.87 mW/m^2 , which suggests a stronger contribution to shortwave (SW from hereafter) scattering by the larger aerosol size distribution, but the large annual model variability makes it difficult to draw any further conclusions.

The annual average atmospheric burden of reentry Al_2O_3 listed in Table 1 is often an order of magnitude larger than the annual burden examined in Ross et al. (2014) for SRMs, yet the net TOA anomaly is on the same scale. The Ross et al. (2014) study only investigated alumina in the accumulation mode size range and confined the aerosols to the 30N–80N latitude band in the lower stratosphere. Furthermore, Ross et al. (2014) performed their calculations without the use of a GCM where atmospheric transport can alter the atmospheric lifetime and exposure of alumina to radiation. The converging of the results presented here with the more idealized Ross et al. (2014) assumptions highlights the influence of atmospheric transport and an evolving aerosol size distribution on reentry alumina's ability to interact with radiation.

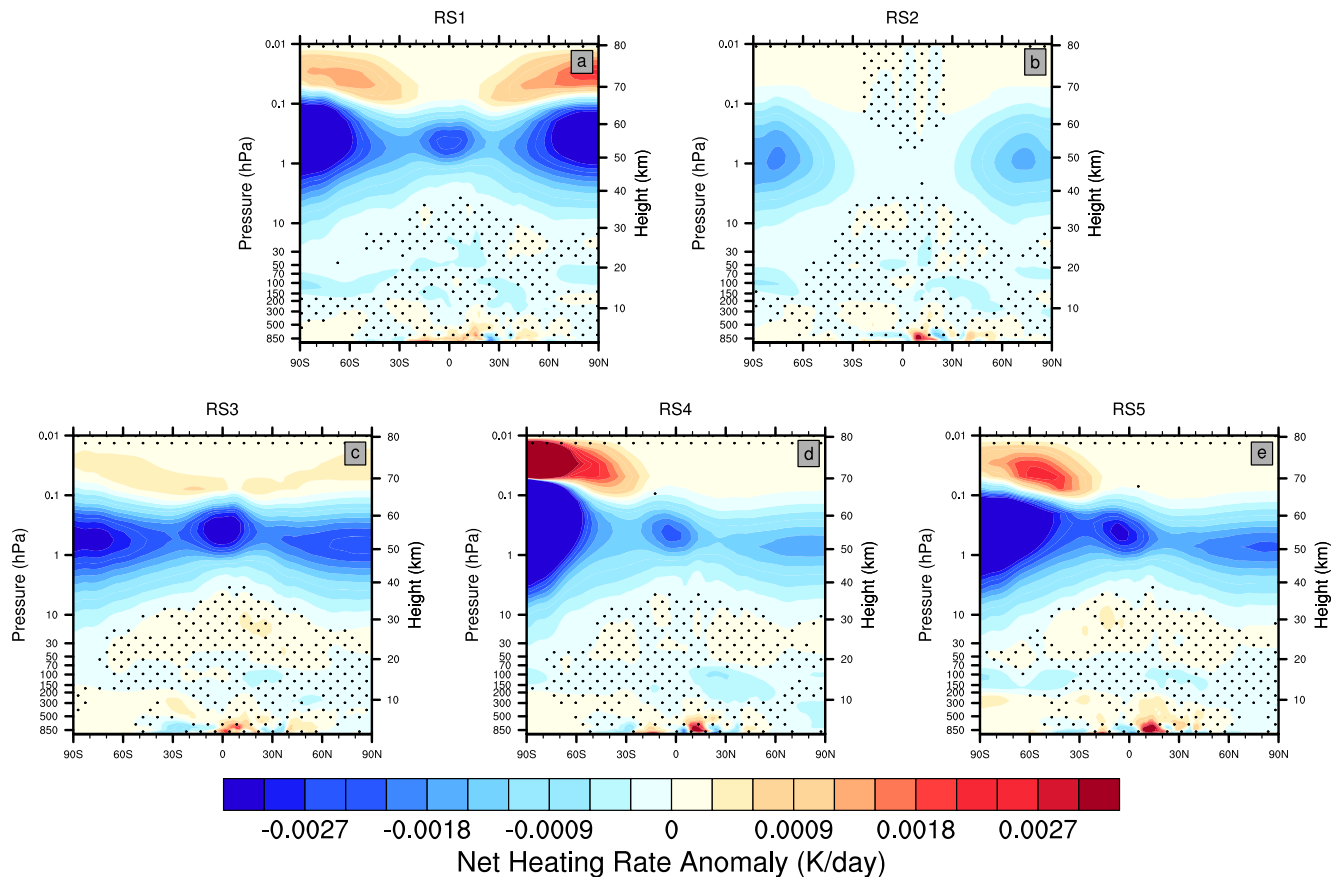


Figure 2. Annual zonal average net heating rate anomaly in K/day for each reentry scenario. The aerosol impact has been isolated by performing a double call to the WACCM radiation code. Statistically significant anomalies (95% confidence interval) lie in areas without the dotted shading.

Another way to estimate the radiative impact from reentry Al_2O_3 is calculating the aerosol-driven change in daily net heating rates (Figure 2). In each scenario, a statistically significant cooling occurs just below the emission altitude band near the stratopause and colocated with the regions of maximum mmr. Directly above the emission regions, heating can be found in all scenarios. This net heating anomaly only appears in the summertime hemisphere (not shown). When one separates out the LW and SW heating rate components (Figure S3 and S4 in Supporting Information S1), it becomes clear that the net heating rate anomaly depicted in Figure 2 is dominated by anomalies in the LW for the RS1, RS3, RS4, and RS5 scenarios. Unlike the other reentry scenarios, the stratospheric net heating rate anomaly in RS2 (Figure 2b) has contributions from both LW and SW heating rate anomalies (Figure S3b and S4b in Supporting Information S1). Overall, the net heating rate anomaly in RS2 is roughly half the strength of the other scenarios. The considerably smaller Al_2O_3 burden in the RS2 stratosphere is likely the cause of this difference (Figure 1).

3.3. The Indirect Temperature Effects of Reentry Al_2O_3

All reentry scenarios produced statistically significant annual temperature anomalies that span from the upper troposphere to the upper mesosphere (Figure 3). The general pattern of the temperature anomaly is consistent in each scenario and is described as follows: Poleward of 50°S and below 50 km, the SH stratosphere and the upper troposphere warms by 0.2–1.6 K. The SH mesosphere experiences a nearly equal in strength cooling poleward of 50°S. In the NH, a cooling of less than 1 K occurs in the upper troposphere/lower stratosphere (UT/LS from herein), as well as in the mesosphere poleward of 60°N. Conversely, the high-latitude NH stratosphere can be seen to warm by approximately 1 K between 20 and 50 km. In all scenarios besides RS1, a mesospheric warming occurs above the stratopause at low latitudes (Figure 3). The differences between each reentry scenario mainly lies in the statistical significance of the temperature anomaly.

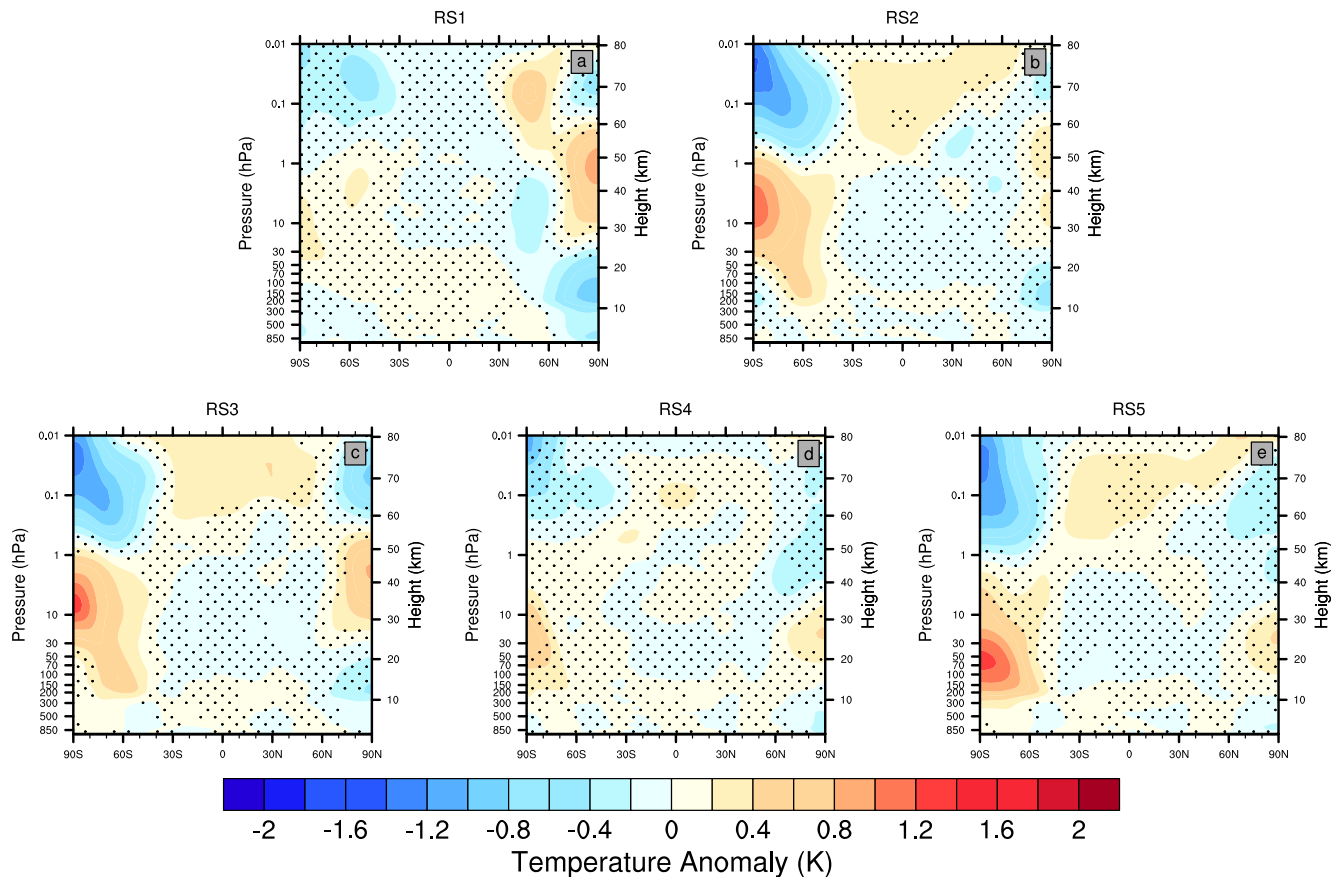


Figure 3. Annual zonal temperature anomalies for each reentry scenario. Statistically significant anomalies (95% confidence interval) lie in areas without dotted shading.

For example, RS1 and RS2, which possess the same reentry pattern but differ in aerosol size distribution, exhibit different patterns of statistical significance (Figures 3a and 3b). In RS1, the NH anomaly is statistically significant, but the SH anomaly is mostly obscured by annual variability. Conversely, the NH RS2 response is not statistically significant, but the SH high-latitude temperature anomalies exceed annual variability (Figure 3b). RS3 and RS5 both show statistically significant high-latitude SH and equatorial mesospheric temperature anomalies (Figures 3c and 3e). RS3 also has a statistically significant high-latitude NH response making it the only reentry scenario to possess a statistically significant annual temperature response in both hemispheres. Conversely, the polar disposal pattern in RS4 rarely exceeds annual variability outside of spatially small high-latitude regions at 25 km (Figure 3d).

3.4. Seasonal and Dynamically Driven Temperature Anomalies

The annual temperature anomaly for RS3 shown in Figure 3c is broken down into monthly averages and further explored in Figure 4. Contour plots of both the SH and NH stratospheric polar vortex latitude band (55 N/S–65 N/S) are included for temperature (top row) and zonal winds (middle row). The monthly mean climatology from the control simulation is overlain as contour lines. The bottom panel in Figure 4 shows the monthly averaged zonal wind time series at the 0.1, 1, and 10 hPa altitudes for each hemisphere. Similar figures for the other reentry simulations can be found in the supplemental information (Figures S5–S8 in Supporting Information S1).

The annual SH temperature anomaly shown in Figure 3 first develops as a cooling of 4 K centered around 0.1 hPa in early April and propagates downward to 1 hPa by the start of the SH springtime (Figure 4a). A stratospheric heating of equal magnitude occurs at the start of May at 10 hPa. This heating persists in November in the lower stratosphere. Accompanying the warming stratosphere is a 10% reduction in the stratospheric polar vortex relative to the mean climatology at 1 hPa that begins in June (Figure 4c). The reduction in upper polar vortex wind speeds persists through September and propagates downward into the lower stratosphere throughout the month of October.

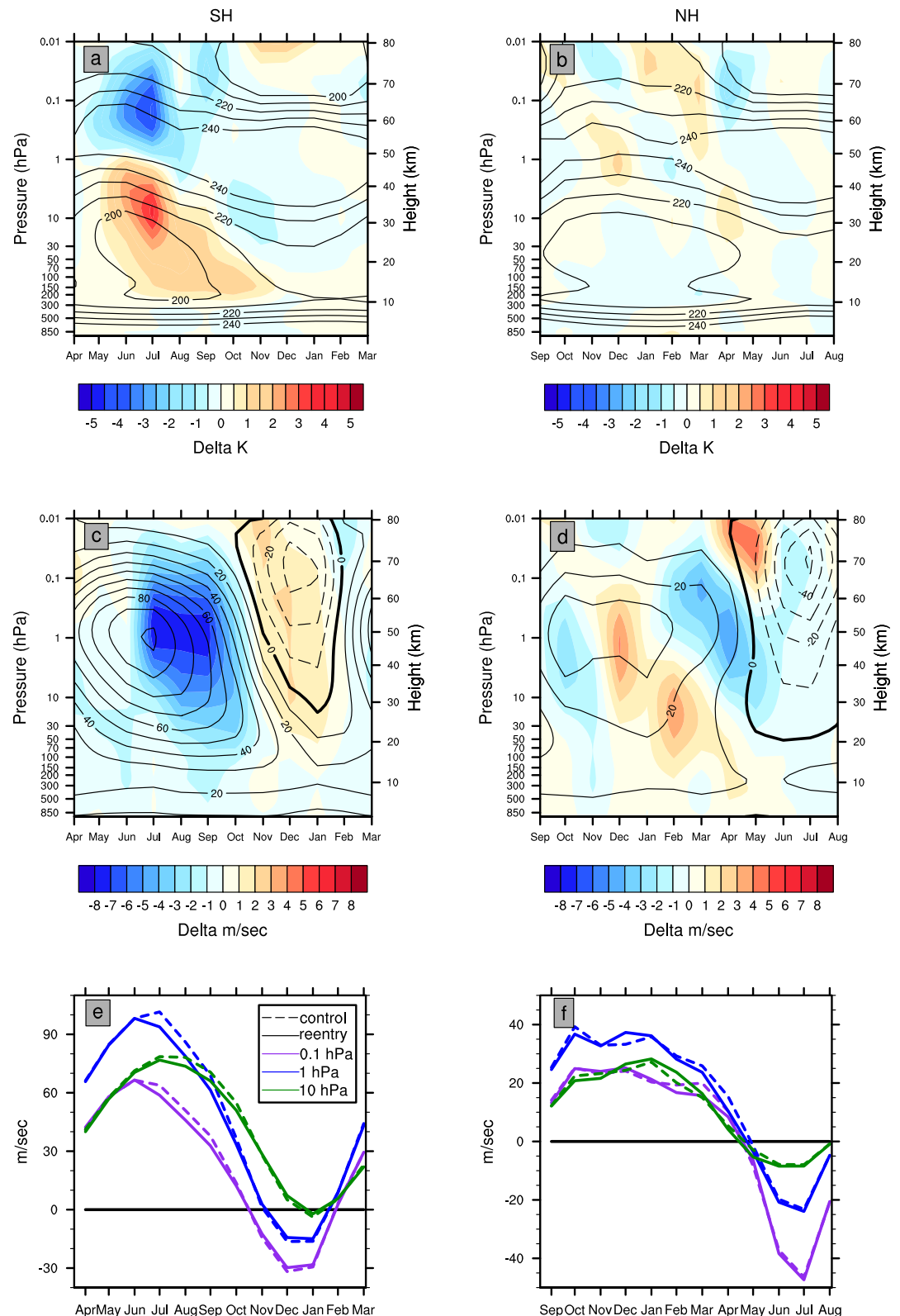


Figure 4. Monthly averaged temperature (a and b) and zonal wind anomalies (c and d) at the 55 S–65 S (left column) and 55 N–65 N (right column) latitude bands for the RS3 reentry scenario. The overlain contour lines represent the monthly mean climatology. Panels (e) and (f) show the monthly mean zonal winds at 0.1 hPa (purple), 1 hPa (blue), and 10 hPa (green) for the reentry simulation (solid line) and control simulation (dashed line).

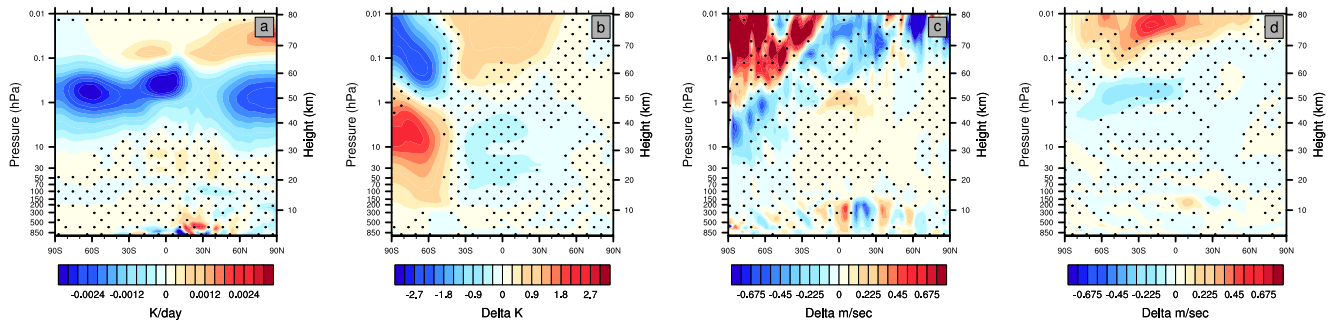


Figure 5. JJA seasonal average for the RS3 reentry scenario. Panel (a) shows the net heating rate anomaly, panel (b) shows the temperature anomaly, panel (c) shows the $\bar{w}^*$ wind residual mean anomaly, and panel (d) shows the $\bar{v}^*$ wind residual mean anomaly. Statistical significance occurs in areas without the dotted shading.

Figure 4e shows that the demise of the SH polar vortex occurs at roughly the same time as the control case suggesting that the zonal wind anomaly is an indication of a weakening winter jet and not a shift to an earlier final warming. RS2, RS3, and RS5 all possess a disruption of the SH polar vortex.

The monthly temperature anomaly in the NH tends to be weaker and is less spatially defined than the SH response (Figure 4b). The NH oscillates between anomalous warming and cooling of 0.5–1 K (Figure 4b). The NH wintertime polar vortex increases in strength by 5% during the December and January months at 1 hPa, and again at 30 hPa in late winter. However, the stronger wintertime zonal winds abruptly weaken during the NH springtime (Figures 4d and 4f).

Scenarios RS1 and RS2 are the only simulations that possess a sustained perturbation of the NH polar vortex (Figure S3 and S4 in Supporting Information S1). Wind speeds increase by ~ 10 m/sec at 10 hPa between January and March in RS1 and for the month of February in RS2. Given that RS1 experiences a statistically significant NH annual temperature response (Figure 3a) while the NH response in RS2 does not exceed the annual model variability (Figure 3b), it is likely that the nearly three times as long stratospheric polar vortex perturbation in RS1 is critical to producing the statistically significant NH stratospheric cooling anomaly seen in Figure 3a.

Figure 5 provides a potential clue to what might be initiating the anomalies in RS2, RS3, and RS5. During the JJA season, no LW heating from reentry Al_2O_3 above 0.1 hPa occurs (Figure 5a) in the SH. Instead, only LW cooling, which maximizes between 0.1 and 1 hPa is present. This cooling in the upper stratosphere/lower mesosphere is enough to trigger more rapid descent between 10 and 1 hPa (Figure 5c). As the air descends, it is warmed via adiabatic heating. Above the heating region at 1 hPa, upward motion is stimulated, which results in the adiabatic cooling of air. Figures 5c and 5d depict this shift in atmospheric transport by showing a decrease in the residual mean circulations for vertical ($\bar{w}^*$) and meridional winds ($\bar{v}^*$). Figure 5c shows a pipeline of increased descent of air from the equatorial mesosphere to the mid-to-high latitude stratosphere. Accompanying this increased downward descent is more robust poleward flow of the deep branch of the Brewer-Dobson circulation near the stratopause (Figure 5d) and a weakening of the mesospheric jet at 30 S in early May (not shown). This JJA SH middle atmospheric temperature response (Figure 5b) and anomalous transport (Figures 5c and 5d) resembles that of a sudden stratospheric warming event (i.e., Andrews et al., 1987; Chandarin et al., 2014; Holton & Alexander, 2000). However, the persistence of this anomalous heating and dynamical shift over the course of the full JJA season suggests a different mechanism for the cause. It is possible that the late fall mesospheric jet disruption at 30 S imposes gravity wave fluctuations that disrupt the normal planetary wave pattern that governs the SH stratospheric polar vortex.

It is difficult to discern the cause of the NH temperature anomalies in RS1 and RS3 depicted in Figure 3. Figure 6 shows DJF seasonal anomalies for the RS1 case. The DJF NH mesosphere warms by as much as 2 K, while the stratosphere cools by an equal amount (Figure 6b). The major differences in the NH response relative to the SH response is a mesospheric equatorial cooling (Figure 6b), an increase in vertical uplift of air in the NH polar stratosphere (Figure 6c), and a slowing of meridional transport in the deep branch of the Brewer-Dobson circulation (Figure 6d). These features are accompanied by an increased isolation of the NH polar stratosphere (Figure S3d in Supporting Information S1). The combination of these anomalies indicate a weaker NH overturning circulation during the DJF season. However, it is not clear why the NH response differs from SH response.

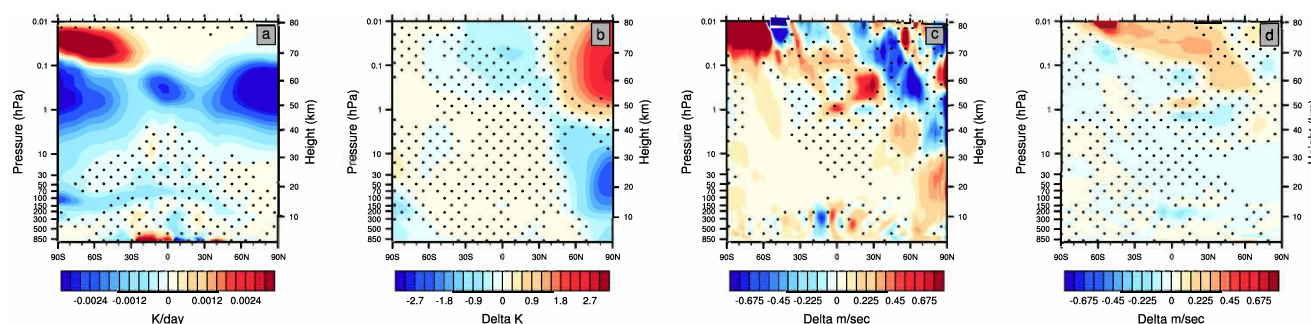


Figure 6. Same as Figure 5 but for the DJF season in the RS1 reentry scenario.

One possible cause could be due to greater amount of wave activity that occurs in NH atmosphere during the DJF season relative to the SH. A more detailed investigation of the NH gravity wave response is beyond the scope of this study.

3.5. Stratospheric Ozone Response

It is clear from the results above that the radiative forcing from Al_2O_3 is small, yet it is still able to trigger shifts in stratospheric dynamics and temperature. One of the major environmental concerns regarding alterations to stratospheric conditions is the effect upon Earth's protective ozone layer.

Figure 7 depicts the annual average zonal ozone anomaly and polar stratospheric cloud fraction anomaly (PSC from hereafter) for the RS3 case. Similar figures for each other reentry scenario can be found in supplemental Figures S9–S12 in Supporting Information S1. The stratospheric ozone response illustrated in Figure 7 (top panel) shows an increase in SH ozone that is colocated with the stratospheric warming in Figure 3c. Ozone increases by 0.2–0.5 DU/km throughout the anomaly region with a maximum column ozone increase of ~ 7 DU within the region. Coinciding with the ozone anomaly, the PSC cloud fraction decreases between 400 hPa and 10 hPa at all latitudes poleward of 60S. The greatest reduction in PSCs occurs at 80 S–90 S with a 3%–6% reduction per model altitude level (Figure 7, bottom panel). Most of the PSC loss occurs during the JJA season while the ozone concentration increases during the SON season. With fewer PSCs present when sunlight returns to the SH high latitude, there is less surface area for ozone destroying chemistry to occur on, thus allowing for higher ozone concentrations to persist at the SH pole.

Conversely, the increased isolation of the NH pole due to a stronger polar vortex leads to colder temperatures and additional PSC formation (Figure 7, bottom panel). Cloud fraction increases at all latitudes poleward of 60 N and altitudes below 100 hPa (Figure 7, bottom panel). The greatest increase in PSC cloud fraction (3%) occurs at 200 hPa, which is colocated with the region of most significant ozone loss in the NH (Figure 7, top panel). A maximum column ozone reduction of ~ 3 DU can be found in the NH region in RS3. We note that when emissions are confined to the SPOUA region in RS5, a ~ 9 DU increase in SH ozone, occurs which is the largest among all reentry scenarios (Figure S10 in Supporting Information S1). RS1 experiences the greatest NH column ozone loss of ~ 5.5 DU (Figure S7 in Supporting Information S1).

4. Discussion

4.1. The Role of Aerosol Size Distribution in the Impact of Al_2O_3

A direct comparison of RS1 and RS2 provides a useful look into how the size distribution of reentry Al_2O_3 influences the climate response. The stratospheric aerosol size distribution for RS1 and RS2 are depicted in Figure 8 for multiple altitude ranges. RS3, RS4, and RS5 all have similar distributions to RS1 and are not shown.

The shape of the distribution does not change with altitude between 10 and 50 km in both scenarios; however, the distribution is more skewed for submicron-sized aerosols between 50 and 70 km. RS1 has a lognormal-shaped distribution that is spread out between 0.02 and 5 μm in diameter. RS2 on the other hand has a bimodal distribution with modes at 0.1 and 20 μm .

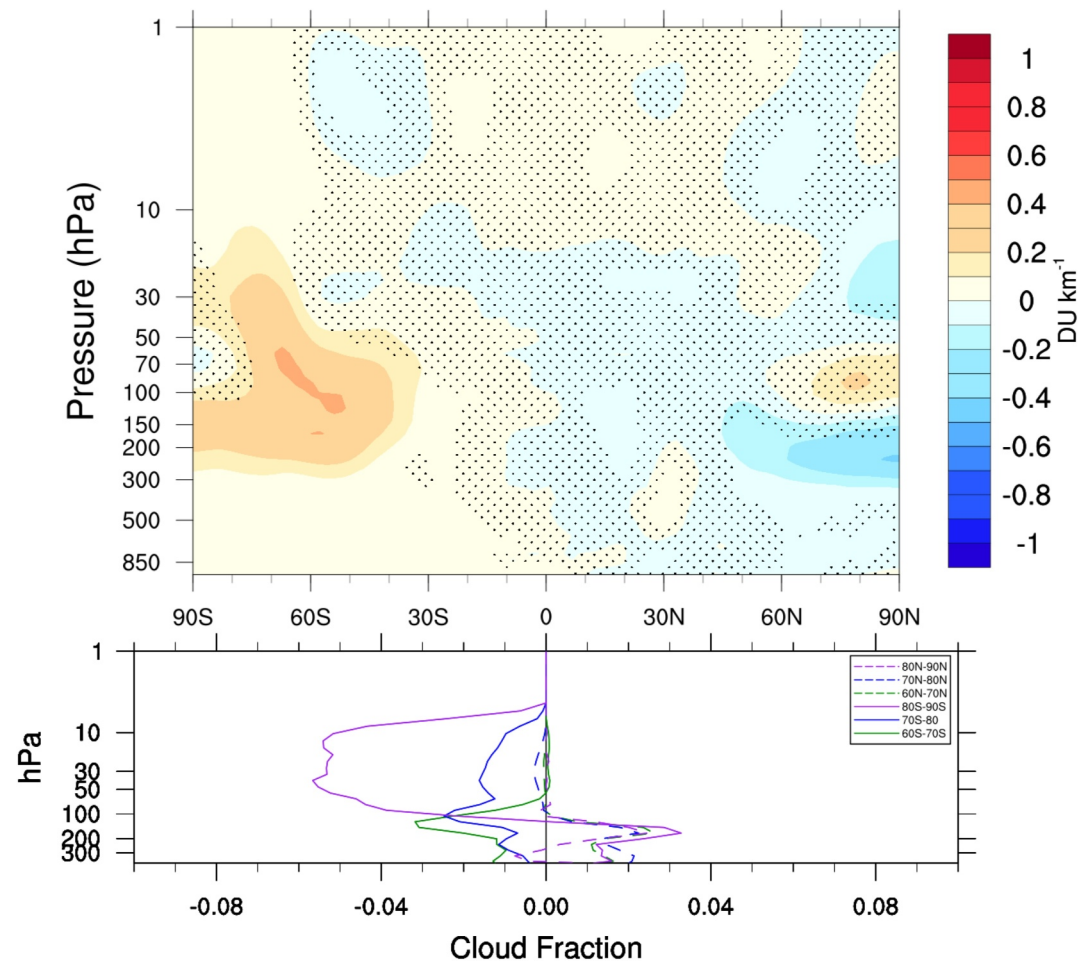


Figure 7. Annual zonal average ozone anomaly (top panel) and cloud fraction anomaly (bottom panel) for RS3. Statistical significance in the ozone anomaly occurs in areas where no dotted shading can be found. The SH and NH cloud fraction anomalies are depicted by the solid and dashed lines, respectively, in the bottom panel. The 60–70 latitude band (green), 70–80 latitude band (blue), and 80–90 latitude band (purple) are shown.

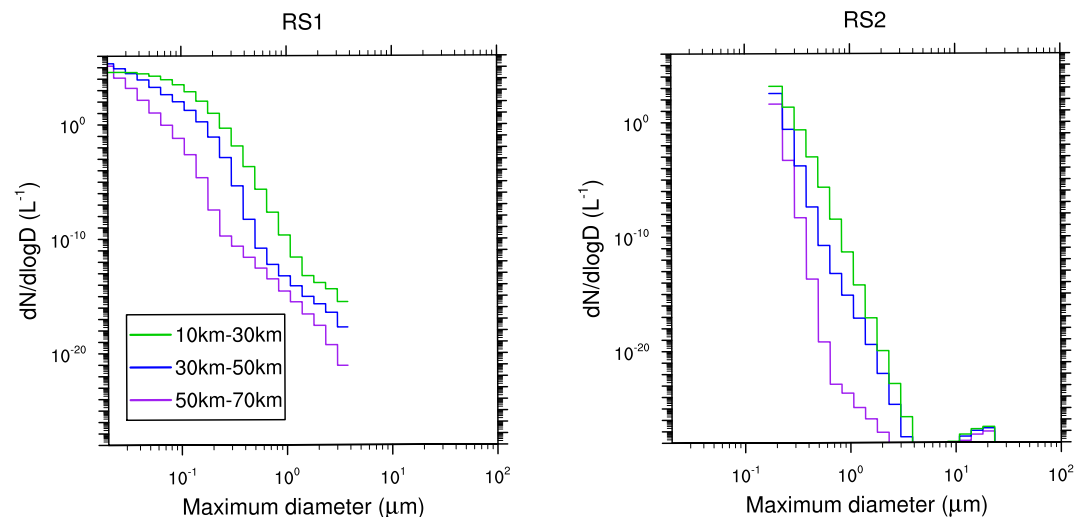


Figure 8. Globally averaged aerosol number density distribution for the RS1 and RS2 reentry scenarios. The 10–30 km- (green), 30–50 km- (blue), and 50–70 km- (purple) altitude ranges are shown.

Though the size distributions in Figure 8 differ considerably in shape and number density, the radiatively driven heating rate anomalies that occurred in both scenarios were still able to produce comparable climate responses. Most notably, RS2 was able to do this with only two-thirds of the stratospheric burden found in RS1. It is likely that the heating rate perturbation causes feedbacks from atmosphere dynamics, temperature, and ozone to occur. The feedbacks can then perpetuate the middle atmosphere response shown above. While it is likely that only a few of these very large aerosols will form per reentry event, our findings suggest that even moderate amounts of these large alumina aerosols can still produce a radiatively driven climate response that is on the same scale as that produced by the smaller aerosol size distribution used in the other scenarios. However, we acknowledge that a mixed reentry aerosol may result in a different response than that shown by the pure Al_2O_3 investigated here.

As previously mentioned, there is currently little known about the production process of reentry aerosol. Multiple pathways are possible with RS1 representing the formation of many small alumina aerosol from the cooling ablation vapor cloud. Conversely, RS2 represents the formation of very large aerosols from rapid coagulation under energetic conditions, or from the shearing off of material from the melting surface of the satellite. Further complicating size distribution is the potential for the reentry aerosol to become mixed with meteoric dust and the background sulfate. Recent work by Vattioni et al. (2024) showed that sulfate may not coat solid aluminum aerosols uniformly, which adds further uncertainty about the composition of these aerosols in the stratosphere.

With all of these unknowns regarding the production of reentry Al_2O_3 , it is clear that additional aircraft in situ observations, as well as laboratory experiments simulating satellite demise with focus upon aerosol composition and size distribution are necessary. Until further measurements are made, we do not possess enough information to truly represent the reentry aerosol size distribution or its radiative response. We can only make rough assumptions as was done in this study. More observations are critical to the ability of further modeling reentry aerosol.

4.2. The Role of Reentry Pattern in the Impact of Al_2O_3

Reentry scenarios RS1, RS3, RS4, and RS5 all used the same aerosol size distribution, but represented different latitude-dependent reentry patterns. Cases RS1 and RS3 both employed a hemispheric symmetrical reentry pattern while disposal in RS4 and RS5 occurred entirely in the SH. Taking into consideration the results presented above, we can begin to form an idea about how differing satellite disposal regimes may influence the climate response. RS3 is the only reentry scenario that experiences a statistically significant climate response in both hemispheres. RS3 also had one of the largest annual stratospheric alumina burdens that also reached high latitudes consistent with the mesospheric meridional transport timeframe. Therefore, it is likely that by emitting reentry Al_2O_3 near the equator, the entire middle atmosphere becomes filled with alumina within a shorter amount of time. This rapid and ubiquitous coverage maximizes reentry alumina's ability to intercept both downwelling solar radiation and upwelling longwave radiation in the middle atmosphere. An equatorial disposal pattern may be the least optimal choice from a radiative impact perspective.

RS4 and RS5, while both occurring entirely in the SH, possess differences in the statistical significance of their climate response. Starting with RS4, it appears that disposing of all reentry material in the SH polar region produces the smallest radiative impact. This is not entirely unexpected as the aerosol is emitted directly into the descending branch of the Brewer-Dobson circulation where air from the middle atmosphere is consistently removed from the stratosphere. Additionally, less solar radiation falls on the polar SH over the course of the year relative to lower latitudes. Therefore, in the RS4 case where reentry Al_2O_3 is directly emitted at polar latitudes, the aerosol has less time to interact with incoming SW radiation. From a radiative transfer and atmospheric loading perspective, a purely polar reentry pattern appears to have the smallest climate impact.

In comparison, RS5 has a more severe temperature response to reentry alumina, as well as a larger disruption to the SH polar vortex (Figure 3 and Figure S6 in Supporting Information S1). It is likely that the specific latitude of the SPOUA region (41 S–46 S) where alumina is emitted in RS5 plays a role. Work by Ray et al. (2017) showed that at most latitudes, air entering the mesosphere in the summertime hemisphere is efficiently transported to the wintertime pole within 6 months. However, air specifically entering between 35 S and 45 S can at times become blocked as it attempts to cross the equator. Once the seasons shift, it is then redirected back toward the SH wintertime pole (Figure 6 from Ray et al., 2017). This atmospheric feature could help explain why RS5 has a larger TOA radiative forcing when compared to the other reentry scenarios as the aerosol-rich air spends a longer time at low SH latitudes (Table 1).

Relative to other reentry disposal patterns, RS1 has a more muted SH response, yet RS1 has the largest impact upon the NH. It is not immediately clear from the results presented here why this is the case. Comparisons to additional climate models other than the WACCM/CARMA, which possess different biases would be useful to further understand this response.

When considering the results presented here, it is important to stress the omission of the potential heterogeneous chemistry that can occur on the surface of the reentry aerosol. Without the inclusion of this chemistry, the surface area for ozone destroying reactions to occur on is not modified. As a result, the ozone response presented here could change, which would have additional downstream effects upon stratospheric heating/cooling and dynamics. Additionally, this study does not investigate the impacts of the sedimentation of Al_2O_3 into the troposphere, which could impact tropospheric cloud formation or polar ice cap albedo. Future studies modeling the impact of reentry aerosols should look to include stratospheric chemistry and the potential tropospheric responses in order to obtain a more complete understanding of the overall climate response.

5. Conclusions

We use the WACCM6/CARMA model to simulate a projected 2040 LEO satellite reentry flux. In these simulations, we assumed that reentry aerosol would be pristine oxidized aluminum. Four different reentry patterns and two aerosol size distributions were examined here. Both aerosol size distribution and the disposal reentry pattern played a significant role in the climate response in each reentry scenario. A completely polar reentry pattern of small aerosols had the smallest climate impact, while lower latitude reentry patterns tended to produce larger, statistically significant responses. Additionally, the larger aerosol produced a similar climate impact to the other reentry scenarios, but with a smaller stratospheric burden.

One important conclusion that can be drawn from this work is that regardless of the reentry disposal pattern and aerosol size distribution, the overall temperature response in the middle atmosphere is consistent across all scenarios with the major differences between each scenario coming from the intensity of the climate response at high latitudes.

This initial modeling study is a first step in a process-based approach to understanding the potential climate impact from a projected increase in satellite reentry mass flux. We find that reentry aerosol can accumulate at high latitudes in the stratosphere within 1–3 years of the start of emissions. Furthermore, the radiative impact is large enough to potentially influence the stratospheric temperature profile, dynamics, and ozone concentration. Therefore, we conclude that it is likely that a 10-Gg/yr emission of Al_2O_3 could impact the middle atmosphere climate. We note that a 10-Gg/yr annual emission of reentry aerosol is a realistic value for a 60,000-LEO satellite population, and is on the conservative end of future projections (i.e., Falle et al., 2023; GAO report 2022).

The results presented here highlight the necessity for expanded investigation into the topic of satellite reentry. Numerous assumptions about the aerosol size distribution, composition, and reentry patterns were made in order to perform this study. Real-world observations are required in order to form a more concrete base of knowledge that is needed to inform future modeling studies. These observations could include, but are not limited to, more high-altitude aircraft in situ observations, which sample the composition and size distribution of reentry aerosol laboratory studies, which simulate the demise of reentering satellites, or remote sensing observations of reentry events.

Additional consideration for other satellite material should also be considered in future modeling work. Other metals and metal oxides including, but not limited to Ti, Li, Fe, and Cu have already been observed to be mixed with present-day stratospheric sulfate (Murphy et al., 2023). Including these additional materials in aerosol modeling is crucial to forward our understanding of the satellite reentry climate impact. Furthermore, a modeling study including heterogeneous ozone chemistry on reentry aerosol is needed and development of that model is currently underway. Future work would also benefit from comparisons between multiple climate models simulating satellite reentry, as well as considering the impact of plume-scale modeling of discrete events (i.e., Danilin et al., 2001) compared to the grid-scale annual emission modeling performed here.

As space technologies advance and become more cost-efficient, continued growth in the space industry is expected. At present, impacts on the middle and the upper atmosphere are small, but have the potential to increase. Emissions from reentering spacecraft ablation could become an important stratospheric aerosol source in the coming decades. Observations of the aerosol processes in this region of the atmosphere are sparse, leaving huge

uncertainties in our estimates of climate impacts. An expanded effort, including observations and modeling is needed in order to shed light upon the potential climate impacts of increased satellite reentry.

Data Availability Statement

WACCM6/CARMA model output from our simulations is publicly available at the University of Colorado's OSF website. The data doi is 10.17605/OSF.IO/KEWV5 (Maloney et al., 2024). The WACCM6 and CARMA models used to perform our simulations fall under version 2.1.3 of the CESM general circulation model, which is publicly available through the National Center for Atmospheric Research webpage (<https://www.cesm.ucar.edu/models>).

Acknowledgments

The work performed here by Dr. Christopher Maloney was supported by the "Modeling of Spaceflight Emissions Impact from 2010 to 2050" NASA Grant (award 80NSSC22K1710), as well as NOAA's Earth's Radiation Budget program. High-performance computing was conducted on the Hera supercomputing system supported by NOAA. The authors would like to thank Dr. Charles Bardeen for his guidance in modifying the CARMA model to run properly on Hera. The authors would also like to thank Dr. Daniel Murphy and Dr. Amy Butler for their helpful discussions.

References

- Ailor, W. (2021). Hazards of reentry disposal of satellites from large constellations. *Journal of Space Safety Engineering*, 6, 2–121. <https://doi.org/10.1016/j.jsse.2019.06.005>
- Ailor, W. (2022). Protecting the LEO environment. *Journal of Space Safety Engineering*, 9–3(3), 449–454. <https://doi.org/10.1016/j.jsse.2022.07.001>
- Andrews, D. G., Holton, J. R., & Leovy, C. B. (1987). *Middle atmosphere dynamics* (No. 40). Academic Press.
- Bardeen, C. G., Toon, O. B., Jensen, E. J., Marsh, D. R., & Harvey, V. L. (2008). Numerical simulations of the three-dimensional distribution of meteoric dust in the mesosphere and upper stratosphere. *Journal of Geophysical Research*, 113(D17), D17202. <https://doi.org/10.1029/2007JD009515>
- Barker, C. R., Marais, E. A., & McDowell, J. (2024). Developing inventories of by-products from satellite megaconstellation launches and disposal to determine the influence on stratospheric ozone and climate. *EGU General Assembly*. <https://doi.org/10.5194/egusphere-egu24-6467>
- Botet, R., Rannou, P., & Cabane, M. (1997). Mean-field approximation of Mie scattering by fractal aggregates of identical spheres. *Applied Optics*, 36(33), 8791–8797. <https://doi.org/10.1364/ao.36.008791>
- Butchart, N. (2014). The Brewer-Dobson circulation. *Reviews of Geophysics*, 52(2), 157–184. <https://doi.org/10.1002/2013RG000448>
- Chandran, A., Collins, R. L., & Harvey, V. L. (2014). Stratosphere-mesosphere coupling during stratospheric sudden warming events. *Advances in Space Research*, 53(9), 1265–1289. <https://doi.org/10.1016/j.asr.2014.02.005>
- Danilin, M. Y., Ko, M. K. W., & Weisenstein, D. K. (2001). Global implications of ozone loss in a space shuttle wake. *Journal of Geophysical Research*, 106(D4), 3591–3601. <https://doi.org/10.1029/2000JD900632>
- Danilin, M. Y., Popp, P. J., Herman, R. L., Ko, M. K. W., Ross, M. N., Kolb, C. E., et al. (2003). Quantifying uptake of HNO₃ and H₂O by alumina particles in Athena-2 rocket plume. *Journal of Geophysical Research*, 108, 4141. <https://doi.org/10.1029/2002JD002601>
- Falle, A., Wright, E., Boley, A., & Byers, M. (2023). One million (paper) satellites. *Science*, 382(6667), 150–152. <https://doi.org/10.1126/science.ad4639>
- Ferreira, J. P., Huang, Z., Nomura, K.-i., & Wang, J. (2024). Potential ozone depletion from satellite demise during atmospheric reentry in the era of mega-constellations. *Geophysical Research Letters*, 51(11), e2024GL109280. <https://doi.org/10.1029/2024GL109280>
- Gardner, C. S., Liu, A. Z., Marsh, D. R., Feng, W., & Plane, J. M. C. (2014). Inferring the global cosmic dust influx to the Earth's atmosphere from lidar observations of the vertical flux of mesospheric NA. *Journal of Geophysical Research: Space Physics*, 119(9), 7870–7879. <https://doi.org/10.1002/2014JA020383>
- Gettelman, A., Mills, M. J., Kinnison, D. E., Garcia, R. R., Smith, A. K., Marsh, D. R., et al. (2019). The Whole Atmosphere Community Climate Model Version 6 (WACCM6). *Journal of Geophysical Research: Atmospheres*, 124(23), 12380–12403. <https://doi.org/10.1029/2019jd030943>
- Government Accountability Office. (2022). *Technology Assessment: Large Constellations of Satellites*. U.S. Government Printing Office.
- Hanning-Lee, M. A., Brady, B. B., Martin, L. R., & Syage, J. A. (1996). Ozone decomposition on alumina: Implications for solid rocket motor exhaust. *Geophysical Research Letters*, 23(15), 1961–1964. <https://doi.org/10.1029/96GL01808>
- Holton, J. R., & Alexander, M. J. (2000). The role of waves in the transport circulation of the middle atmosphere. *Geophysical Monograph Series*, 123, 21–35. <https://doi.org/10.1029/gm123p0021>
- Jones, H. (2018). The recent large reduction in space launch cost. In *48th International Conference on Environmental Systems ICES-2018-81*.
- Kärräng, P., Lips, T., & Soares, T. (2019). Demisability of critical spacecraft components during atmospheric re-entry. *Journal of Space Safety Engineering*, 6(3), 181–187. <https://doi.org/10.1016/j.jsse.2019.08.003>
- Larson, E. J. L., Portmann, R. W., Rosenlof, K. H., Fahey, D. W., Daniel, J. S., & Ross, M. N. (2017). Global atmospheric response to emissions from a proposed reusable space launch system. *Earth's Future*, 5(1), 37–48. <https://doi.org/10.1002/2016EF000399>
- Li, F., Waugh, D. W., Douglass, A. R., Newman, P. A., Strahan, S. E., Ma, J., et al. (2012). Long-term changes in stratospheric age spectra in the 21st century in the Goddard Earth Observing System Chemistry-Climate Model (GEOSCCM). *Journal of Geophysical Research*, 117(D20), D20119. <https://doi.org/10.1029/2012JD017905>
- Linz, M., Plumb, R., Gerber, E., Haenel, F. J., Stiller, G., Kinnison, D. E., et al. (2017). The strength of the meridional overturning circulation of the stratosphere. *Nature Geoscience*, 10(9), 663–667. <https://doi.org/10.1038/ngeo3013>
- Maloney, C. M., Portmann, R. W., Ross, M. N., & Rosenlof, K. H. (2022). The climate and ozone impacts of black carbon emissions from global rocket launches. *Journal of Geophysical Research: Atmospheres*, 127(12), e2021JD036373. <https://doi.org/10.1029/2021JD036373>
- Maloney, C. M., Portmann, R. W., Ross, M. N., & Rosenlof, K. H. (2024). Investigating the potential atmospheric accumulation and radiative impact of the coming increase in satellite reentry frequency. <https://doi.org/10.17605/OSF.IO/KEWV5>
- Marsh, D. R., Mills, M. J., Kinnison, D. E., Lamarque, J., Calvo, N., & Polvani, L. M. (2013). Climate change from 1850 to 2005 simulated in CESM1 (WACCM). *Journal of Climate*, 26(19), 7372–7391. <https://doi.org/10.1175/jcli-d-12-00558.1>
- Mills, M., Toon, O., Turco, R., Kinnison, D., & Garcia, R. (2008). Massive global ozone loss predicted following nuclear conflict. *Proceedings of the National Academy of Sciences of the United States of America*, 105(14), 5307–5312. <https://doi.org/10.1073/pnas.0710058105>
- Murphy, D., Abou-Ghanem, M., Cziczo, D., Froyd, K., Jacquot, J., Lawler, M., et al. (2023). Metals from spacecraft reentry in stratospheric aerosol particles. *Proceedings of the National Academy of Sciences of the United States of America*, 120(43), e2313374120. <https://doi.org/10.1073/pnas.2313374120>
- Pardini, C., & Anselmo, L. (2020). Environmental sustainability of large satellite constellations in low earth orbit. *Acta Astronautica*, 170, 27–36. <https://doi.org/10.1016/j.actaastro.2020.01.016>

- Pardini, C., & Anselmo, L. (2024). The risk of casualties from the uncontrolled re-entry of spacecraft and orbital stages. *Journal of Space Safety Engineering*, 11(2), 181–191. <https://doi.org/10.1016/j.jsse.2024.02.002>
- Park, S.-H., Navarro Laboulais, J., Leyland, P., & Mischler, S. (2021). Re-entry survival analysis and ground risk assessment of space debris considering by-products generation. *Acta Astronautica*, 179, 604–618. <https://doi.org/10.1016/j.actaastro.2020.09.034>
- Park, S.-H., & Park, G. (2017). Reentry trajectory and survivability estimation of small space debris with catalytic recombination. *Advances in Space Research*, 60(5), 893–906. <https://doi.org/10.1016/j.asr.2017.05.004>
- Plane, J. M. C. (2012). Cosmic dust in the earth's atmosphere. *Chemical Society Reviews*, 41(19), 6507–6518. <https://doi.org/10.1039/C2CS35132C>
- Plane, J. M. C., Daly, S. M., Feng, W., Gerding, M., & Gomez Martin, J. C. (2021). Meteor-ablated aluminum in the mesosphere-lower thermosphere. *Journal of Geophysical Research: Space Physics*, 126(2), e2020JA028792. <https://doi.org/10.1029/2020JA028792>
- Ray, E. A., Moore, F. L., Elkins, J. W., Rosenlof, K. H., Laube, J. C., Röckmann, T., et al. (2017). Quantification of the SF₆ lifetime based on mesospheric loss measured in the stratospheric polar vortex. *Journal of Geophysical Research: Atmospheres*, 122(8), 4626–4638. <https://doi.org/10.1002/2016JD026198>
- Ross, M. N., Benbrook, J. R., Sheldon, W. R., Zittel, P. F., & McKenzie, D. L. (1997). Observations of stratospheric ozone depletion in rocket exhaust plumes. *Nature*, 390(6655), 62–64. <https://doi.org/10.1038/36318>
- Ross, M. N., & Sheaffer, P. M. (2014). Radiative forcing caused by rocket engine emissions. *Earth's Future*, 2(4), 177–196. <https://doi.org/10.1002/2013EF000160>
- Ryan, R. G., Marais, E. A., Balhatchet, C. J., & Eastham, S. D. (2022). Impact of rocket launch and space debris air pollutant emissions on stratospheric ozone and global climate. *Earth's Future*, 10(6), e2021EF002612. <https://doi.org/10.1029/2021EF002612>
- Schulz, L., & Glassmeier, K.-H. (2021). On the anthropogenic and natural injection of matter into Earth's atmosphere. *Advances in Space Research*, 67(3), 1002–1025. <https://doi.org/10.1016/j.asr.2020.10.036>
- Toon, O. B., Pollack, J. B., & Khare, B. N. (1976). The optical constants of several atmospheric aerosol species: Ammonium sulfate, aluminum oxide, and sodium chloride. *Journal of Geophysical Research*, 81(33), 5733–5748. <https://doi.org/10.1029/JC081i033p05733>
- Toon, O. B., Turco, R. P., Westphal, D., Malone, R., & Liu, M. S. (1988). A multidimensional model for aerosols: Description of computational analogs. *Journal of the Atmospheric Sciences*, 45(15), 2123–2143. [https://doi.org/10.1175/1520-0469\(1988\)045<2123:AMMFAD>2.0.CO;2](https://doi.org/10.1175/1520-0469(1988)045<2123:AMMFAD>2.0.CO;2)
- Tropf, W. J., & Thomase, E. T. (1997). Aluminum Oxide (Al₂O₃) revisited. In E. D. Palik (Ed.), *Handbook of optical constants of solids* (Vol. III, pp. 653–682). Academic Press. <https://doi.org/10.1016/B978-012544415-6.50124-2>
- Vattioni, S., Weber, R., Feinberg, A., Stenke, A., Dykema, J. A., Luo, B., et al. (2024). A fully coupled solid particle microphysics scheme for stratospheric aerosol injections within the aerosol-chemistry-climate-model SOCOL-AERv2. *EGU sphere*. <https://doi.org/10.5194/egusphere-2024-444>
- Waugh, D. W., & Hall, T. M. (2002). Age of stratospheric air: Theory, observations, and models. *Reviews of Geophysics*, 40(4), 1010. <https://doi.org/10.1029/2000RG000101>
- Weisenstein, D. K., Keith, D. W., & Dykema, J. A. (2015). Solar geoengineering using solid aerosol in the stratosphere. *Atmospheric Chemistry and Physics*, 15(20), 11835–11859. <https://doi.org/10.5194/acp-15-11835-2015>